TORSION FREENESS SURVIVES BASE CHANGE ALONG A FLAT HOMOMORPHISM OF DOMAINS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Torsion element and torsion free module over a domain. definition environment at source lines 4781-4792 of the same section. Supplies the elementwise meaning of the hypothesis on M and of the conclusion on M tensor R'.
- A module is torsion free if and only if M to M tensor K is injective, K the fraction field. the discussion printed at source lines 4794-4799, immediately before the counted statement. Converts torsion freeness of M into the injection M to M tensor K that is then tensored with R'.
- Flatness of R over R' makes the functor tensor R' preserve injectivity. used at proof lines 4832-4833, from the flatness hypothesis in the statement at source lines 4824-4826. Keeps the injection after tensoring with R', which is the step that makes the whole reduction possible.
- Localisation of a domain is torsion free over that domain. used at proof lines 4835-4836 for K tensor R'. Identifies the remaining tensor product as a localisation, settling the final case.

Source reference. Original lemma 0AXM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13760–13760. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $R \to R'$ be a flat homomorphism of domains. If M is a torsion free R-module, then $M \otimes_R R'$ is a torsion free R'-module.*

Proof. If M is torsion free, then $M \subset M \otimes_R K$ is injective where K is the fraction field of R. Since R' is flat over R we see that $M \otimes_R R' \to (M \otimes_R K) \otimes_R R'$ is injective. Since $M \otimes_R K$ is isomorphic to a direct sum of copies of K, it suffices to see that $K \otimes_R R'$ is torsion free. This is true because it is a localization of R'. $\square$